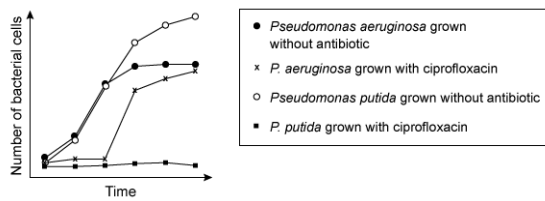


Two species of bacteria, *Pseudomonas aeruginosa* and *Pseudomonas putida*, were incubated separately for 24 hours in broth with or without the antibiotic ciprofloxacin. Afterward, a small portion of each group was transferred to fresh broth without ciprofloxacin, and bacterial growth was measured for another 24 hours, as shown in the graph below.



Which statement is best supported by the data in the graph?

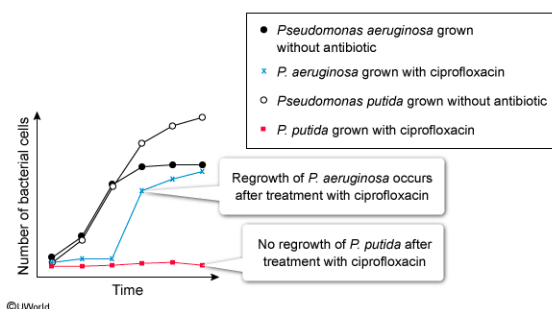
- ☐ A. Treatment with no antibiotic results in a greater rate of regrowth in *P. aeruginosa* than *P. putida*. [19%]
- ☐ B. Regrowth of *P. aeruginosa* treated with ciprofloxacin occurs more slowly than regrowth of *P. putida* treated with ciprofloxacin. [6%]
- ☐ C. Treatment with ciprofloxacin results in faster regrowth than treatment with no antibiotic in *P. aeruginosa*. [24%]
- ☒ D. Regrowth of *P. aeruginosa*, but not *P. putida*, occurs slowly after treatment with ciprofloxacin. [49%]

Correct

49% answered correctly

Explanation:

Bacterial regrowth after antibiotic treatment



The growth of a bacterial population occurs when bacteria undergo **binary fission** to reproduce. The rate at which this occurs can be demonstrated by plotting a typical **growth curve** that includes the following phases: lag, exponential growth, stationary, and death. This growth curve can change when bacteria are incubated with an antibiotic (ie, a substance that inhibits growth).

In this example, two species of bacteria (*Pseudomonas aeruginosa* and *Pseudomonas putida*) were grown either with or without the antibiotic ciprofloxacin for 24 hours. Afterward, a small portion of each group was transferred to fresh broth that did not contain ciprofloxacin, and bacterial growth was measured for an additional 24 hours.

The graph shows that in this regrowth experiment, only *P. aeruginosa* initially grown with

ciprofloxacin demonstrates regrowth over time following transfer to antibiotic-free broth. In contrast, *P. putida* initially grown with the same antibiotic does *not* show regrowth during this time period. Therefore, **regrowth of *P. aeruginosa***, but **not *P. putida***, occurs slowly after initial treatment with **ciprofloxacin**.

(Choice A) Not adding antibiotic initially results in a greater rate of regrowth of *P. putida* compared to *P. aeruginosa*.

(Choice B) Regrowth of *P. putida* initially grown with ciprofloxacin does *not* occur within this time period, but regrowth of *P. aeruginosa* initially treated with ciprofloxacin *does* occur. Therefore, regrowth of *P. aeruginosa* treated with ciprofloxacin occurs more quickly, *not* more slowly, than regrowth of *P. putida* treated with ciprofloxacin.

(Choice C) Initial treatment of *P. aeruginosa* with *no* antibiotic results in faster regrowth than initial treatment with ciprofloxacin.

Educational objective:

The bacterial growth curve includes the following phases: lag, exponential growth, stationary, and death. Accordingly, the rate at which populations of bacteria grow typically follows this curve unless an antibiotic is introduced, which may disrupt population growth.

Time spent:0 sec

Subject:
Biology

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System:
Cellular Biology